

Western New England University
College of Engineering
ECE Department
Wave Transmission and Reception
EE 457/557
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Design Project #4
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Design Project #4

Element	Score	Max
Design		500
HFSS		300
Tables		100
Presentation		100
Overall		1000

Design Project 04

14-GHz Waveguide Resonant Longitudinal Slot Array Antenna

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EE-457 Wave Transmission and Reception

Western New England University



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Design Objectives:

- Design an 8-element ($N_y = 8$) waveguide resonant longitudinal slot array antenna at 14 GHz
- Waveguide: WR-62 ($a = 0.622''$, $b = 0.311''$)
- Top wall thickness: $t = 25$ mils = 0.635 mm
- Remaining walls: $t_2 = 40$ mils = 1.016 mm

Design Specifications:

- Element spacing: $d_y = \lambda_g/2$
- Slot dimensions: $L = 0.464\lambda_0$, $W = \lambda_g/20$
- Chebyshev synthesis with -26 dB SLL
- One-Dimensional (1D) array (1×8) and Two-Dimensional (2D) array (8×8)

Design Parameters:

f_0	14 GHz
a	15.7988 mm
b	7.8994 mm
λ_g	29.121 mm
d_y	14.5605 mm
L	9.9360 mm
W	1.4560 mm

1 × 8 Array Model

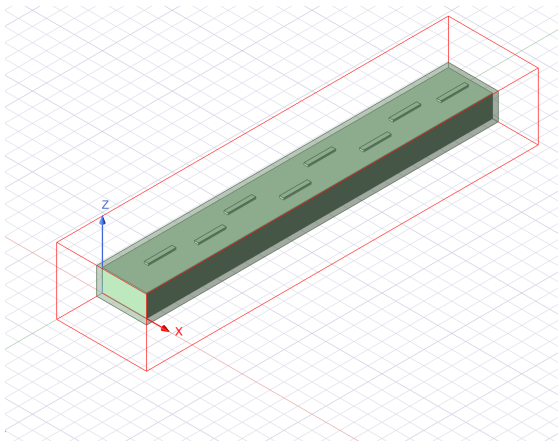


Figure 1: HFSS model of the 1 × 8 element waveguide slot array antenna.

8 × 8 Array Model

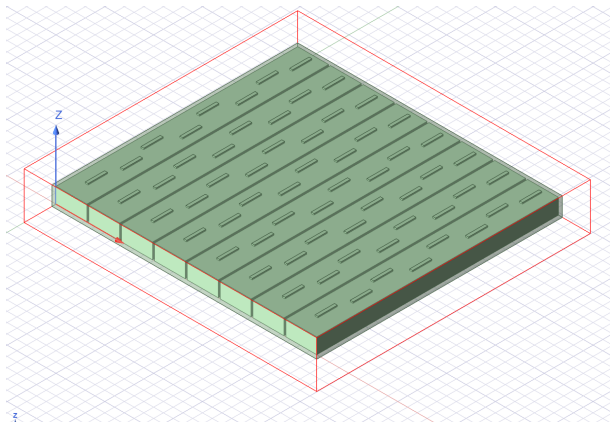


Figure 2: HFSS model of the 8 × 8 element waveguide slot array antenna.

Table 1: Chebyshev Amplitude Coefficients



A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8
1	1.6313	2.3916	2.8603	2.8603	2.3916	1.6313	1

Table 1: Amplitude for each element in the 14-GHz waveguide resonant longitudinal slot array antenna synthesized employing the Chebyshev method with a -26 dB side lobe level.

Design Note

The Chebyshev amplitude distribution provides symmetric tapering with maximum amplitude at the center elements (A_4, A_5), achieving the specified -26 dB sidelobe level while maintaining reasonable beamwidth.

Table 2: Design Parameters for Slot Array



A_n	g_n (mS/S)	L_n (mm)	W_n (mm)	Δ_n (mm)
1	28.4700	9.9360	1.4560	-0.9726
1.6313	75.7611	9.9360	1.4560	1.6038
2.3916	162.8404	9.9360	1.4560	-2.4019
2.8603	232.9286	9.9360	1.4560	2.9272
2.8603	232.9286	9.9360	1.4560	-2.9272
2.3916	162.8404	9.9360	1.4560	2.4019
1.6313	75.7611	9.9360	1.4560	-1.6038
1	28.4700	9.9360	1.4560	0.9726

Table 2: Table 2: Summary of the design parameters for the 14-GHz waveguide resonant longitudinal slot array antenna.

Slot Offset Pattern

The alternating signs of Δ_n place slots on opposite sides of the waveguide centerline, ensuring proper phasing for broadside radiation.

Table 3: HFSS Simulated Response



Table 3: Table 3: Summary of the simulated response (HFSS) for the 14-GHz waveguide resonant longitudinal slot array antenna.

Parameter	1 × 8 Element Array	8 × 8 Element Array	Unit
G	34.70	272.42	W/W
G	15.40	24.35	dB
BW_{EP}	11.24	11.26	◦
BW_{HP}	77.69	9.72	◦

Design Observations

The 8×8 array (8×8) achieves approximately 9 dB higher gain than the 1D array, with significantly narrower H-plane beamwidth due to the 8-element aperture in that dimension.

1D Array: $|S_{11}|_{(\text{dB})}$ vs $\theta^{(\circ)}$ Gain Pattern (Polar)

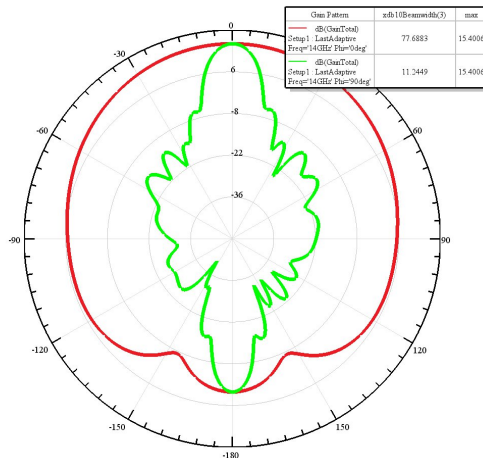


Figure 3: Polar gain pattern for the 1×8 element slot array at 14 GHz. E-plane (green, $\phi = 90^\circ$) and H-plane (red, $\phi = 0^\circ$) cuts shown with 10 dB beamwidth markers.

1D Array: $|S_{11}|_{\text{(dB)}} \text{ vs } \theta(^{\circ})$ Gain Pattern

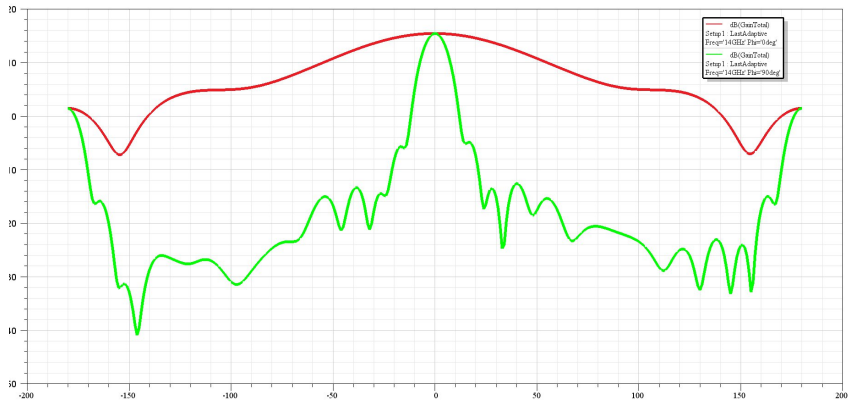


Figure 4: Rectangular gain plot for the 1×8 element array at 14 GHz showing E-plane (green) and H-plane (red) patterns.

1D Array: $|S_{11}|_{\text{(dB)}} \text{ vs } f_{\text{(GHz)}}$

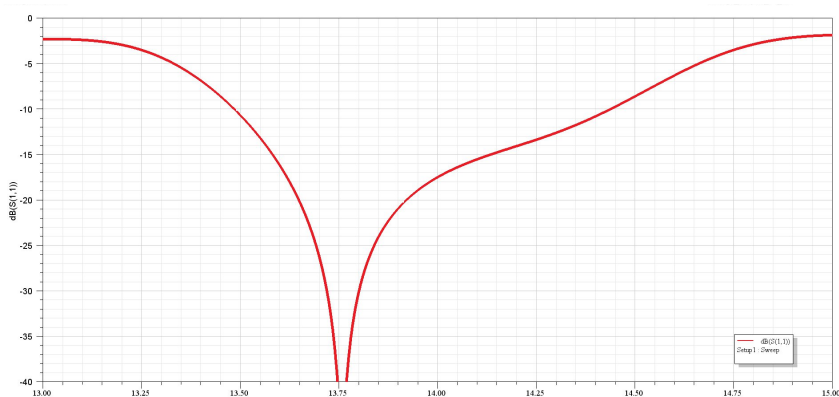


Figure 5: $|S_{11}|$ (dB) vs frequency for the 1×8 element array. The resonance occurs near 13.75 GHz with $|S_{11}| < -40$ dB. 🚀

8×8 Array: $|S_{11}|_{(\text{dB})}$ vs $\theta(^{\circ})$ Gain Pattern (Polar)

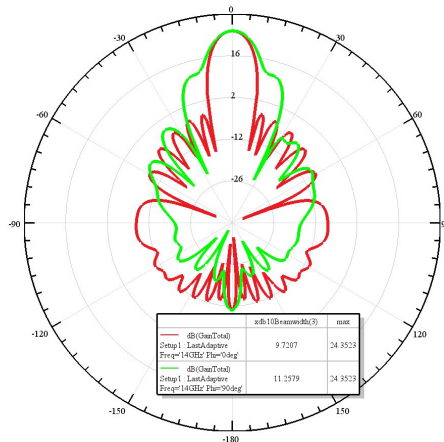


Figure 6: Polar gain pattern for the 8×8 element slot array at 14 GHz. Peak gain of 24.35 dB with narrow beamwidths in both principal planes.

8×8 Array: $|S_{11}|_{\text{(dB)}} \text{ vs } \theta^{(\circ)}$ Gain Pattern

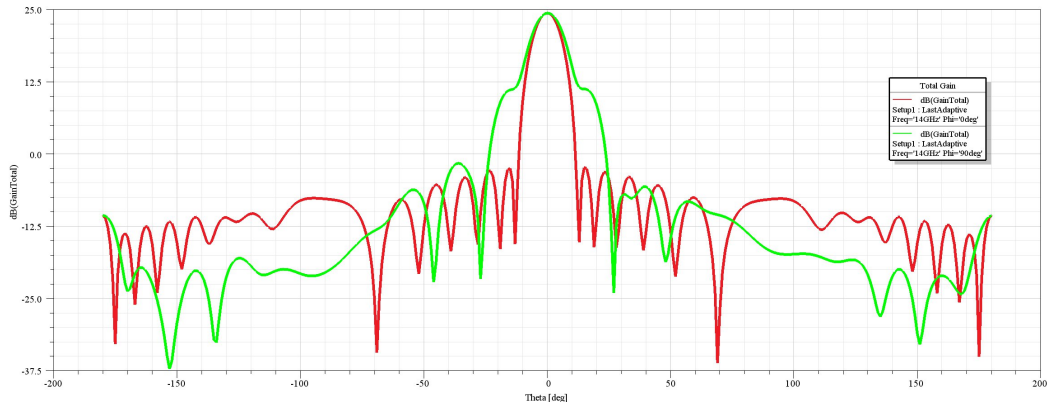


Figure 7: Rectangular gain plot for the 8×8 element array at 14 GHz showing the principal plane cuts.

8×8 Array: 3D Radiation Pattern

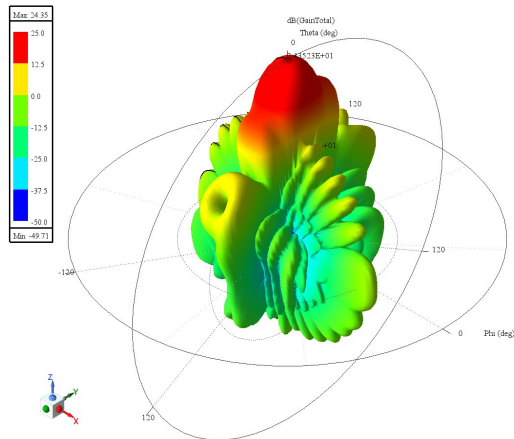


Figure 8: 3D radiation pattern for the 8×8 element array showing the pencil beam at broadside with peak gain of 24.35 dB.

8×8 Array: S-Parameters

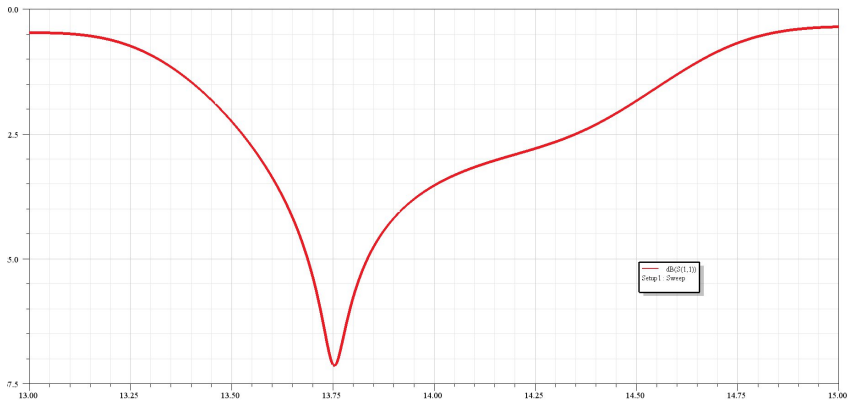


Figure 9: $|S_{11}|_{\text{(dB)}}$ vs $f_{\text{(GHz)}}$ for the 8×8 element array showing the input match.

8×8 Array: All Port S-Parameters

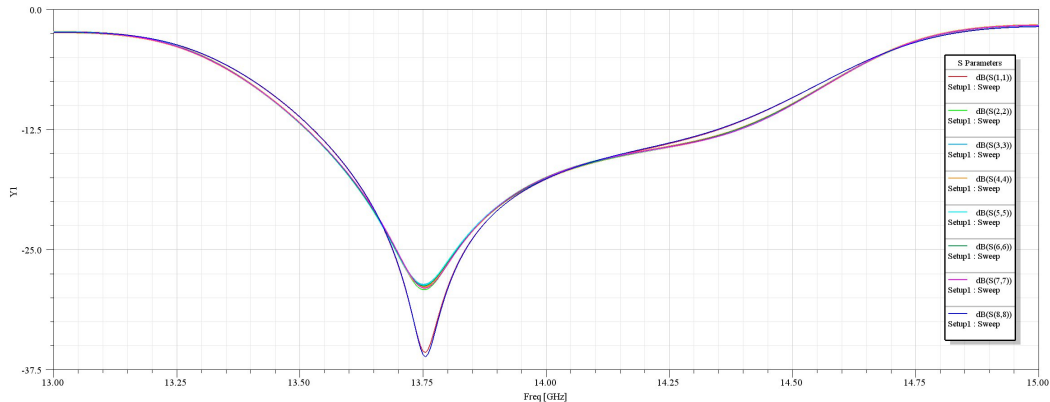


Figure 10: S-parameters ($|S_{(1,1)}|_{\text{(dB)}}$ thru $|S_{(8,8)}|_{\text{(dB)}}$ vs $f_{\text{(GHz)}}$) for all 8 waveguide ports of the 8×8 array. All ports show similar resonance behavior near 13.75 GHz.



Accomplishments:

- ① Successfully designed a 14-GHz waveguide resonant longitudinal slot array antenna
- ② Implemented Chebyshev amplitude tapering for -26 dB sidelobe level
- ③ Simulated both 1×8 and 8×8 element configurations in HFSS
- ④ Achieved target gain and beamwidth specifications

Results:

- 1D Array: $G = 15.40$ dB, $BW_{EP} = 11.24^\circ$, $BW_{HP} = 77.69^\circ$
- 8×8 Array: $G = 24.35$ dB, $BW_{EP} = 11.26^\circ$, $BW_{HP} = 9.72^\circ$
- Good input match achieved at all ports ($|S_{nn}| < -25$ dB at resonance)

Design Validation

The HFSS simulations confirm the Chebyshev synthesis methodology produces the expected sidelobe performance and gain characteristics for waveguide slot arrays.



References:

Ryan Leonard, Aidan Butler, Tommy Nguyen, Fevin Farjado

Software Tools:

- Ansys HFSS 2024 R1 for electromagnetic simulation of antenna arrays
- MATLAB for Chebyshev synthesis and parameter calculation
- $\text{\LaTeX}$ for presentation formatting

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Appendix I – Additional HFSS Plots

- 1D Array: Gain Pattern
- 8×8 Array: Gain Pattern
- 8×8 Array: 3D Gain Pattern

Appendix II – MATLAB Code

- Lines 1–20: Units & Design Parameters
- Lines 21–40: WR-62 Dimensions & Specs
- Lines 41–55: Waveguide & Slot Factors
- Lines 56–64: File Locations
- Lines 65–80: Chebyshev Weights
- Lines 81–88: 1D Array Factor
- Lines 89–100: Guided Wavelength
- Lines 100–118: Slot Admittance Model
- Lines 119–135: HFSS Geometry
- Lines 136–159: Write HFSS Parameters
- Lines 160–177: Write Arrays to File
- Lines 178–192: Write Excel Tables



Appendix I

Additional HFSS Plots

Gain Patterns for 1D and 2D Arrays

1D Array: Gain Pattern

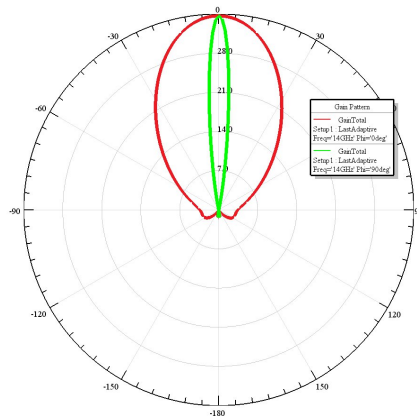


Figure 11: Gain pattern (polar) for the 1 × 8 element array at 14 GHz.

8 × 8 Array: Gain Pattern

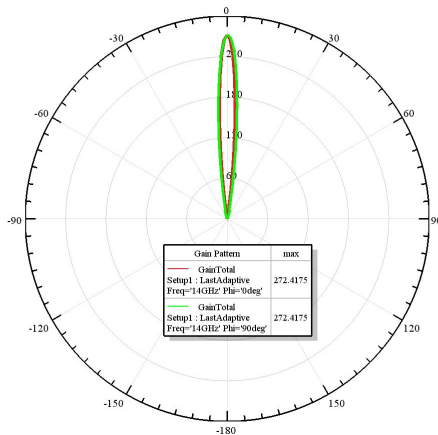


Figure 12: Gain pattern (polar) for the 8 × 8 element array at 14 GHz showing maximum gain of 272.4175 (linear).

8 × 8 Array: 3D Linear Gain Pattern

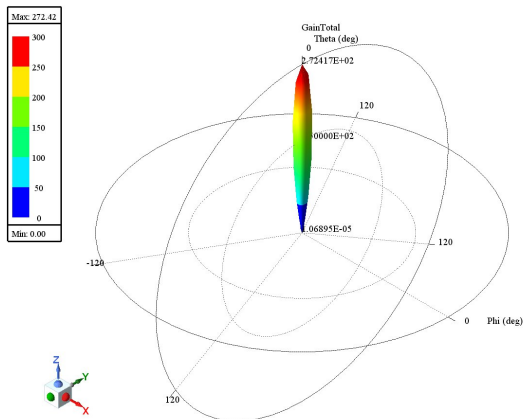


Figure 13: 3D radiation pattern (linear scale) for the 8 × 8 element array showing the highly directive pencil beam.



Appendix II

MATLAB Design Code

EE_457_Design_Project_04_Placzek.m

Lines 1–192

EE-457
Wave Tx and Rx
Design Project 04

Lines 1 – 40



```
1 % EE-457 Design Project 04 - Slotted Waveguid Array Design
2 % Steven Placzek
3
4 clearvars;
5 close all;
6 clc;
7 j = 1j;
8
9 %% Units
10 G = 1e9; % Hz per GHz
11 m = 1e-3; % 1 mm in meters (used for m <-> mm conversion)
12
13 %% Design parameters (Project 4: 14 GHz, WR-62, 8-element
14 % Chebyshev, -26 dB SLL)
15 f0_GHz = 14.000; % design frequency in GHz
16 f0 = f0_GHz * G; % design frequency in Hz
17 uc = 299792458; % uc = speed of light (m/s)
18 u0 = 1.25663706212*10^-6; % H/m
19 e0 = 8.8541878128*10^-12; % F/m
20 q = 1.602176634*10^-19; % C
21 eta_0 = 376.730313668; % Ohms
```

```
21 % WR-62 internal dimensions (broad x narrow) in inches
22 % (converted below)
23 a_imp_in = 0.622; % WR-62 broad wall (in)
24 b_imp_in = 0.311; % WR-62 narrow wall (in)
25
26 D_lambda = 1/2; % synthesis spacing d/lambda0 for AF
27 Ny = 8; % number of slots per waveguide
28 Nx = 8; % number of waveguides
29
30 SLL_dB = 26; % |SLL| magnitude in dB (spec is -26 dB)
31 theta_MB = 0; % main beam angle (deg)
32 phi = 90; % E-plane (array along y, slots in x)
33 NF = 32;
34 IFigure = 0;
35
36 % Wall and slot thickness (from Project 4 spec)
37 topWallThick_mm = 0.635; % 25 mils top wall & slot thickness
38 sideWallThick_mm = 1.016; % 40 mils remaining walls
39
40 %}
```



```
41 topWallThick = topWallThick_mm * m; % [m]
42 sideWallThick = sideWallThick_mm * m; % [m]
43 t = topWallThick;
44 t2 = sideWallThick;
45
46 % Waveguide dimensions (meters)
47 X_in_to_cm = 2.54;
48 X_cm_to_m = 1e-2;
49 X_in_to_m = X_in_to_cm * X_cm_to_m;
50 a = a_imp_in * X_in_to_m; % waveguide broad wall [m]
51 b = b_imp_in * X_in_to_m; % waveguide narrow wall [m]
52
53 % Slot length and width factors (per spec: L = 0.464*lambda0, W
54   = (1/20)*lambda_g)
55 slot_length_factor = 0.464;
56 slot_width_factor = 1/20;
```

```
56 % File locations (user-specific)
57 baseDir = pwd; % or a project root
58 File_Loc_Excel = ...
59     fullfile(baseDir, ...
60     'EE_457_Design_Project_04_Placzek_Excel.xlsx');
61 Sheet = 'Table_01';
62 File_Loc_Txt = ...
63     fullfile(baseDir, ...
64     'EE_457_Design_Project_04_Placzek_Parameters.txt');
```



```
65 %% -----  
66 % Chebyshev weights (Ny elements, -26 dB SLL)  
67 %% -----  
68 Np = Ny - 1;  
69 M = 0.5*Ny - 0.5*mod(Ny, 2); % M = Ny/2 for even Ny  
70 kd = 360 * D_lambda; % kd in degrees (d =  
    D_lambda*lambda0)  
71 beta = -kd * sind(theta_MB) * sind(phi);  
72 % Amplitude ratio R = |AF|_max / |AF|_sidelobe for Chebyshev  
73 R = 10^(SLL_dB/20);  
74 u0 = cosh((1/Np) * acosh(R));  
75 [T_N, uzi] = EE457_Chebyshev_Poly(Np); %#ok<ASGLU>  
76 psi_zi = 2 * acosd(uzi / u0);  
77 wzi = Polar_2_Rect(1, psi_zi);  
78 An = real(poly(wzi));  
79 % Normalize so that A1 = 1.0 (matches Table 1 normalized  
    amplitudes)  
80 An = An / An(1);
```

```
81 % 1D array factor (normalized, for verification of SLL, not used  
    by HFSS)  
82 theta = (-90:0.5:90).';  
83 psi = kd * sind(theta) * sind(phi) + beta;  
84 w = Polar_2_Rect(1, psi);  
85 AF = polyval(An, w);  
86 AFn = AF ./ max(abs(AF));  
87 AFn_Mag = abs(AFn);  
88 AFn_dB = 20*log10(AFn_Mag); %#ok<NASGU>
```



```
89 %% -----  
90 % Guided wavelength and normalized target conductances  
91 %% -----  
92 fc      = uc / (2*a);           % TE10 cutoff frequency  
93 lambda0 = uc / f0;             % free-space wavelength at f0  
94 X_f      = sqrt(1 - (fc/f0)^2);  
95 lambda_g = lambda0 / X_f;       % guided wavelength  
96  
97 % Normalize power: sum(gn) = 1.0  
98 K = 1 / sum(An.^2);  
99 gn = K .* An.^2;               % target normalized conductances  
    g_n
```

```
100 %% -----  
101 % Slot dimensions and slot admittance model  
102 %% -----  
103 L = slot_length_factor * lambda0 * ones(size(An)); % slot  
    length [m]  
104 W = slot_width_factor * lambda_g * ones(size(An)); % slot  
    width [m]  
105  
106 Cg = cos(pi * L ./ lambda_g);  
107 C0 = cos(pi * L ./ lambda0);  
108 S0 = sin(pi * L ./ lambda0);  
109  
110 slotConst = 480 / (73*pi); % TE10 slot admittance constant  
111 T1 = slotConst;  
112 T2 = lambda_g / lambda0;  
113 T3 = a / b;  
114 T4 = ((Cg - C0) ./ S0).^2;  
115  
116 Xg = T1 * T2 * T3 * T4; % normalized slot susceptance  
117 delta = (-1).^(1:Ny) .* (a/pi) .* asin(sqrt(gn ./ Xg));  
118 g = Xg .* sin(pi*delta/a).^2; %ok<NASGU>
```

EE-457
Wave Tx and Rx
Design Project 04

119 – 159



```
119 %% -----  
120 % Waveguide dimensions and array spacing (for HFSS geometry)  
121 %% -----  
122 dx = a + sideWallThick; % center-to-center spacing of  
    waveguides (x)  
123 dy = 0.5 * lambda_g; % slot spacing along waveguide (y)  
124 L_WG = Ny*dy + 0.5*dy; % resonant waveguide length (end  
    short at ~lambda_g/4)  
125 bw = b + t + t2; % outer box b-dimension (z)  
126 aw_2D = Nx*dx + t2; % outer box a-dimension for Nx  
    waveguides  
127 aw_1D = dx + t2; % outer box a-dimension for single  
    waveguide  
128 % Slot locations along the top wall (HFSS coordinates)  
129 xs = 0.5*a + delta - 0.5*W; % slot start x (top wall)  
130 ys = ((1:Ny).' * dy) - 0.5*L(1); % slot start y (vectorized)  
131 % Outer PEC box for HFSS  
132 xb = -t2;  
133 yb = -t2;  
134 zb = -t2;  
135 Lb = L_WG + 2*t2;
```

```
136 %% -----  
137 % Write HFSS parameter text file (dimensions in mm)  
138 %% -----  
139 FID_txt = fopen(File_Loc_Txt, 'wt');  
140 if FID_txt < 0  
141     error('Could not open parameter file: %s', File_Loc_Txt);  
142 end  
143 fprintf(FID_txt, 'a = %.4f \n', a/m);  
144 fprintf(FID_txt, '\nb = %.4f \n', b/m);  
145 fprintf(FID_txt, '\nL_WG = %.4f \n', L_WG/m);  
146 fprintf(FID_txt, '\nL = %.4f \n', L(1)/m);  
147 fprintf(FID_txt, '\nW = %.4f \n', W(1)/m);  
148 fprintf(FID_txt, '\nt = %.4f \n', t/m);  
149 fprintf(FID_txt, '\ndx = %.4f \n', dx/m);  
150 fprintf(FID_txt, '\ndy = %.4f \n', dy/m);  
151 fprintf(FID_txt, '\nxb = %.4f \n', xb/m);  
152 fprintf(FID_txt, '\nyb = %.4f \n', yb/m);  
153 fprintf(FID_txt, '\nzb = %.4f \n', zb/m);  
154 fprintf(FID_txt, '\naw_1D = %.4f \n', aw_1D/m);  
155 fprintf(FID_txt, '\naw_2D = %.4f \n', aw_2D/m);  
156 fprintf(FID_txt, '\nbw = %.4f \n', bw/m);  
157 fprintf(FID_txt, '\nLb = %.4f \n', Lb/m);  
158 fprintf(FID_txt, '\nt2 = %.4f \n', t2/m);  
159 fprintf(FID_txt, '\ndelta = [0]');
```



```
160 for kk = 1:length(delta)
161     fprintf(FID_txt, ', %.4f', delta(kk)/m);
162 end
163 fprintf(FID_txt, ', 0] \n');
164
165 fprintf(FID_txt, '\nxs = [0');
166 for kk = 1:length(xs)
167     fprintf(FID_txt, ', %.4f', xs(kk)/m);
168 end
169 fprintf(FID_txt, ', 0] \n');
170
171 fprintf(FID_txt, '\nys = [0');
172 for kk = 1:length(ys)
173     fprintf(FID_txt, ', %.4f', ys(kk)/m);
174 end
175 fprintf(FID_txt, ', 0] \n');
176
177 fclose(FID_txt);
```

```
178 %% -----
179 % Write Excel design tables (Table 1 / Table 2 style)
180 %% -----
181 ni = (1:(Ny - 1)).';
182 M_D = [ni, uzi, psi_zi];
183 writematrix(M_D, File_Loc_Excel, "Sheet", Sheet, "Range", 'A3');
184
185 M_D = An;
186 writematrix(M_D, File_Loc_Excel, "Sheet", Sheet, "Range", 'F3');
187
188 M_D = [An; gn/m; delta/m];
189 writematrix(M_D, File_Loc_Excel, "Sheet", Sheet, "Range", 'F6');
190
191 M_D = [An; gn/m; L/m; W/m; delta/m];
192 writematrix(M_D, File_Loc_Excel, "Sheet", Sheet, "Range",
            'A16');
```




Thank You

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